

IMPORTS

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

SEABORN BIVARIATE

```
sns.scatterplot(x='total_bill', y='tip', data=tips)
sns.regplot(x='total_bill', y='tip', data=tips)
sns.regplot(x='total_bill', y='tip', data=tips,
            fit_reg=False)
sns.jointplot(x='total_bill', y='tip', data=tips)
sns.jointplot(x='total_bill', y='tip', data=tips,
              kind='hex')
sns.jointplot(x='total_bill', y='tip', data=tips,
              kind='kde')
sns.barplot(x='time', y='total_bill', data=tips)
sns.barplot(x='time', y='total_bill', data=tips,
            estimator='std')
sns.boxplot(x='time', y='total_bill', data=tips)
sns.violinplot(x='time', y='total_bill', data=tips)
```

LOAD DATA

```
anscombe = sns.load_dataset('anscombe')
tips = sns.load_dataset('tips')
```

SEABORN UNIVARIATE

```
sns.histplot(tips['total_bill'])
sns.histplot(tips['total_bill'], kde=True)
sns.kdeplot(tips['total_bill'])
sns.rugplot(tips['total_bill'])
sns.countplot(x='day', data=tips)
```

STYLES

```
sns.set_style('whitegrid')
sns.set_style('darkgrid')
sns.set_style('dark')
sns.set_style('white')
sns.set_style('ticks')
```

FIGURE / AXES

```
fig = plt.figure()
ax = fig.add_subplot(1, 1, 1)
fig, ax = plt.subplots()
fig, axes = plt.subplots(2, 2, figsize=(8, 8))
axes[0, 0].plot(...)
```

MULTIVARIATE

```
sns.violinplot(x='time', y='total_bill', hue='sex',
               data=tips, split=True)
sns.lmplot(x='total_bill', y='tip', data=tips,
           hue='sex', fit_reg=False)
sns.lmplot(..., markers=['o','x'], scatter_kws={'s':
           tips['size'] * 10})
```

DO NOT USE

```
sns.distplot(...)
```

AX METHODS

```
ax.plot(x, y)
ax.plot(x, y, 'o')
ax.set_title('Title')
ax.set_xlabel('x')
ax.set_ylabel('y')
fig.suptitle('Title')
fig.tight_layout()
plt.show()
```

FACETS

```
sns.lmplot(x='x', y='y', data=anscombe, fit_reg=False,
           col='dataset', col_wrap=2)
sns.lmplot(x='total_bill', y='tip', data=tips,
           fit_reg=False, hue='sex', col='day')
facet = sns.FacetGrid(tips, col='time')
facet.map(sns.histplot, 'total_bill')
facet = sns.FacetGrid(tips, col='day', hue='sex')
facet.map(plt.scatter, 'total_bill', 'tip')
facet.add_legend()
facet = sns.FacetGrid(tips, col='time', row='smoker',
                      hue='sex')
sns.catplot(x='day', y='total_bill', hue='sex',
            data=tips, row='smoker', col='time', kind='violin')
```

MATPLOTLIB PLOTS

```
ax.hist(tips['total_bill'], bins=10)
ax.scatter(tips['total_bill'], tips['tip'])
ax.boxplot(..., labels=...)
ax.scatter(x=..., y=..., s=..., c=..., alpha=0.5)
```

PAIRWISE

```
sns.pairplot(tips)
sns.pairplot(tips, hue='sex')
grid = sns.PairGrid(tips)
grid.map_upper(sns.scatterplot)
grid.map_lower(sns.kdeplot)
grid.map_diag(sns.histplot)
```

PANDAS PLOT

```
df['col'].plot()
df['col'].plot.hist()
df.plot.scatter(x='a', y='b')
df.plot.box()
df.plot.bar()
df.plot.line()
df.plot.area()
df.plot.density()
df.plot.hexbin(x='a', y='b')
```